

AI in Business Decision Making: Transforming Organisational Intelligence in the Digital Era

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ABSTRACT

Artificial Intelligence (AI) has emerged as a transformative force in contemporary business ecosystems, fundamentally reshaping how organisations collect, analyse, and act upon information to make strategic decisions. This paper investigates the multifaceted role of AI in business decision-making across operational, managerial, and strategic levels with a focus on real-world adoption patterns between 2020 and 2025. Using a mixed-methods research design combining a quantitative survey of 280 business executives with qualitative case analyses of five leading organisations, the study reveals that AI-driven decision-making tools significantly enhance accuracy, speed, and cost-efficiency. Organisations integrating AI into their core decision frameworks report a 38.4% improvement in decision accuracy and a 31.7% reduction in decision-cycle time. However, key barriers including algorithmic bias, data privacy concerns, workforce resistance, and infrastructure limitations persist. The paper proposes a structured AI Decision Integration Framework (ADIF) as a roadmap for sustainable AI adoption. Three research hypotheses are tested and validated through Structural Equation Modelling

Keywords: Artificial Intelligence; Decision Support Systems; Organisational Intelligence; Digital Strategy; Executive Analytics

1. The Evolution of Enterprise Decision Intelligence

Algorithmic decision-support systems have progressed from rudimentary descriptive analytics dashboards to prescriptive, self-optimizing neural networks. Decision speed has become a key competitive differentiator across fast-evolving modern markets.

2. Empirical Survey of 280 Corporate Executives

Data indicates that 71.4% of surveyed enterprises utilize machine learning for customer churn prediction and inventory forecasting. However, governance deficits remain the leading cause of algorithmic decision abandonment.

3. The AI Decision Integration Framework (ADIF)

The ADIF articulates four stages: foundational data hygiene, human-in-the-loop pilot testing, enterprise-wide workflow integration, and continuous ethical audits for bias prevention.

4. Conclusion & Corporate Recommendations

Executive leadership must champion algorithmic explainability and data democratization. AI should be positioned as cognitive augmentation rather than wholesale autonomous human replacement.

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